

Virus B-gone!

Viruses are by far the most visible reason of security. Having the advantage of being among the first threats to affect computers, viruses have had that much more time to grab hold of the world's attention. Despite being so well-known (or perhaps because of it), many myths abound about what viruses and their lesser cousins are, and what they're capable of.

Although there's no hard and fast definition, it's generally agreed that a virus is a piece of code that has the ability to self-replicate—that is, it can make copies of itself automatically, once it enters a host computer. The code doesn't have to do anything nasty to be called a virus. This is something a lot of people aren't aware of. In a way, the very fact that the program is multiplying in your computer without your knowledge can be termed as malicious activity.

To be or not to be...

Although viruses for other platforms have been around for decades now, most agree that the first virus to spread from PC to PC was Brain. Legend has it that in the mid-1980s, Basit and Amjad Farooq Alvi, owners of Brain Computer Services in Lahore, wrote the boot sector virus to ascertain the extent of software piracy in Pakistan (they were software vendors). But Brain soon leaked through the Pakistani borders and quite harmlessly infected computers worldwide. Since then, there have been thousands of viruses added to the list.

According to IBM researcher Sarah Gordon, author of a number of books examining the mindset of virus authors, there are an estimated 8,000 to 10,000 virus writers in the world—the vast majority of who are teenage boys. The truth is that almost all these virus creators simply take existing viruses and modify a few bytes to create what are called variants—giving them bragging rights in the process. That's right; most viruses are the result of pre-pubescent insecurity. Only once in a while does a truly original creation come along and disrupt our collective existence.

Over the years, all this activity has led to the creation of several types of malicious code. Digging deeper into this world, you'll come across the following terms related to programs that cause you much pain.

Viruses: File Viruses attach themselves to selected programs, employing various techniques to do so. They are spread whenever an infected program or file is used.

Resident File Viruses stay in the computer's memory after the files they reside in are called up. Thus, each time a program is executed, the infection is repeated. This type of virus is harder to eliminate and some can even survive a 'soft reboot'. These are the most common types of viruses.

Stealth Viruses were developed to go undetected by virus scanners—they use a camouflage mechanism which conceals their existence.

Polymorphic Viruses are the most dangerous type of virus. This category consists of viruses which change their appearance with every new infection. They normally do this by re-encoding their virus signature with a random key, which alters from infection to infection.

System Viruses use components, such as the master boot record, partition table and boot sector, as the infection medium.



They spread whenever the affected disk or media is used. For example, Boot Sector Viruses attack the boot sector on floppy disks, or hard disks. The infection spreads slowly as infection can only take place when the system is booted from an infected disk.

Trojans: Coined by former hacker Dan Edwards, and referring to the Trojan Horse used in the mythical battle of Troy, these programs are disguised as something benign; such as a directory lister, archiver, game—and in one notorious case in 1990 on the Mac, a program to find and destroy viruses! These programs open a backdoor, through which anyone with the right tools can take complete control of your PC.

Worms: Taken from 'tapeworm' in John Brunner's novel *The Shockwave Rider*, these are programs that propagate themselves over a network, reproducing as they go. Perhaps the best-known example is Robert T. Morris' Great Worm of 1988, a benign worm that got out of control, and brought the Internet to its knees.

The Virus Museum

Let's take a few well-known recent examples of such programs to see their modus operandi:

Nimda

According to CERT.org's warning, released on September 18th 2001, the worm modifies Web documents (.htm and .asp files), and certain executable files found on a system, and creates numerous copies of itself under various file names. This worm propagates through e-mail containing an attachment named 'readme.exe', which is a binary executable. Due to a vulnerability in Internet Explorer-based e-mail clients, the enclosed attachment is executed and, as a result, infects the machine. Additionally, it creates numerous copies of itself in all writable directories, (including those found on a network share) to which the user has access.

Klez

This worm and virus combination apparently originated from Asia—possibly China, or Hong Kong. The first infections were